

Research article

The influence of service performance in China's sci-tech commissioner system: Using social network analysis and interpretable machine learning

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ABSTRACT

The Sci-Tech Commissioner System (SCS) is a result of exploratory efforts by the Chinese government to use science and technology to strengthen the agricultural sector. Social network analysis (SNA) and machine learning (ML) techniques make it feasible to assess the service performance in China's SCS by using indicators such as group types and structure features. In this study, SNA and a clustering algorithm were employed to categorize service group types of sci-tech commissioners. By comparing the accuracy of different classification algorithms in predicting the clustering results, LightGBM algorithm was finally select to determine the clustering features of sci-tech commissioners and establish an interpretable ML model. Then, the SHAP was used to algorithm to analyze influences affecting service performance. Results show that the service forms of sci-tech commissioners are group-oriented, and that group types include small groups of young commissioners with close cooperation, larger groups of young and middle-aged commissioners, small groups of middle-aged and old commissioners with close cooperation, and isolated points of highly-influential commissioners. Furthermore, while group size is not the determinant of a commissioner's average performance, group structure and coordination ability were found to be more critical. Moreover, while differences in distinct types of service performance are caused by various factors, but good group structures and extensive social contacts are essential for high service performance.

1. Introduction

There has been a global movement of establishing technical institutions in rural regions, and thereby transitioning technical personnel into these areas to carry out science and technology development. For example, there has been a movement in the United States to prioritize the involvement of land-grant universities in scientific and technological services development and delivery, particularly in researching new techniques and providing training [1]. Similarly, many developing countries are extending their agricultural services with the aim to expedite technological adoption and subsidize national production, such as the Farm Input Subsidy Programme in Malawi and the Rural Capacity Building Project in Ethiopia [2,3]. Research shows that designating specialists to provide farmers with financial and technical assistance helps to address problems in agricultural production, facilitates the transfer of

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agricultural technology, and accelerate resource flow, all of which help to improve agricultural productivity and innovation levels overall [4,5].

In China, the Sci-Tech Commissioner System (SCS) is one such program encouraging exploratory scientific and technological research to empower agriculture. Through the SCS, various levels of government are deploying scientific or technological experts from universities and research institutions into rural areas to conduct a research and measurements while also providing agricultural producers in these areas with access specialized guidance. The impact of the SCS on rural revitalization in China has been noticeable. Since 2002, Zhejiang Province alone has dispatched 39,000 researchers and specialists, resulting in a provincial increase in gross farmers' income of 6.353 billion yuan and an improvement in industrial performance of 4.51 billion yuan.¹ However, the effectiveness of the SCS in achieving sustainable development depends upon the service performance of the provisional services provided, which reflects the overall operational efficiency of the regional public sector, the quality of the sci-tech services provision being provided, and the extent to which these services meet the expectations of the service recipients, that is, the agricultural workers and individual enterprises [6–8].

Current literature on the service performance of sci-tech support is focused on the identification of specific relationships, exploration of effects, and the evaluation of particular methods of knowledge or technology services being transferred. Several studies have demonstrated that conventional variables considered in assessing public management, such as civil servant wages, internal enforcement of rules and regulations, and agency autonomy, as well as external factors such as public participation and public-private partnerships, all play vital roles in bolstering service performance [9–11]. Moreover, some studies have emphasized the link between high-performance sci-tech assistance and the levels of farmers' risk aversion or skepticism towards new technologies, stressing the significance of clear information and the availability of consultancy support in increasing the dissemination of scientific and technological advancements across all industry participants [12–14]. However, much of this research has been centered on the development of performance evaluation indicators, especially static indicators and a priori judgement, capturing the macro perspective of policy formulation, industrial development, risk challenges, and evolutionary process [5,15,16]. In contrast, synergies arising out of complex cooperative networks have received less attention, and few scholars have focused on the potential impact of service staff interactions on service performance [17,18]. Specific to the SCS, research findings have indicated that greater attention should be given to the interactions and relationships among internal stakeholders, and more consideration should be given to continuity in collaborations [19,20]. Meanwhile, in terms of research methods, most studies have adopted linear analyses, such as index comparison or multilinear regression, which can to some extent limit the possibility of in-depth explorations [16,21].

Based on the limitation of the current literature, as mentioned above, this study aimed to comprehensively measure the service performance of the SCS by incorporating commissioner service data and collaboration networks. More specially, this study focused on the individuals who make up the SCS service network to explore group types and structures from a micro-aspect. Furthermore, to better understand service performance of the SCS, social network analysis (SNA) was applied to extract information about the social interaction networks following terms of specific rules they follow and their modes in order to examine interplays between various nodes within these complex cooperation networks [22,23]. Considering the remarkable accuracy of machine learning (ML), its ability to handle high dimensional data, and its access to computing power to excavate non-linear relationships [24], an ML algorithm was employed to reveal the driving factors of the service performance of the SCS. In addition to the ML algorithm, we also adopted SHAP (SHapley Additive exPlanations), an interpretable tool, which can illuminate active mechanisms at work by providing a visual comparison of results and their varied patterns [25].

This study makes the following contributions to the existing literature. First, this study enriches the theoretical understandings of the SCS service performance by combing SNA and ML to classify the different members involved into various group types based on their interactions, while also utilizing ML to investigate the influence of diverse structural characteristics on SCS service performance. Second, this study employed ML to identify the complicated mechanisms at work within SCS service performance, as well as their influencing patterns, which compensates for limitations in sample size and the use of traditional linear methods. Third, this study provides practical recommendations as to how our methodology and results can be used to clarify commissioner selection protocols and improve existing government cooperation schemes, which may promote resource integration and optimize the organizational structure of the SCS overall.

2. Literature review

2.1. China's sci-tech commissioner system (SCS)

The introduction of the SCS in several regions of China in 2002 was intended to stimulate the coordination and collaboration between sci-tech specialists and innovators with relevant government departments at various levels, and to stimulate the integration of modern production elements into rural areas, in terms of technology, information, management, and capital.² The goals of the SCS have been threefold: (1) providing farmers with improved agricultural technology; (2) facilitating the exchange of knowledge experience while imparting new technologies; and (3) the development of online information management platforms [26,27].

Since its inception, researchers in China have paid close attention to SCS service performance. From the macro perspective, efforts have been made to compile indicators which assess service performance in various provinces. For example, Huang et al. suggested the

¹ <http://zj.people.com.cn/n2/2023/0824/c186327-40543327.html>.

² http://english.www.gov.cn/policies/latest_releases/2016/05/19/content_281475352888461.htm.

adoption of five primary indicators - sci-tech development projects, communities of interest, service to farmers, technology extension, and poverty alleviation - and utilized an improved entropy weight method to macroscopically compare and contrast the service performance of different provinces [28]. Conversely, other research has adopted a micro perspective to emphasize SCS service performance within a single province. For instance, using the AHP-entropy weight method, Li et al. obtained service data from Fujian Province collected between 2019 and 2021 and stored on a cloud platform, and suggested that policy advocacy, sci-tech dissemination, innovation and entrepreneurship, and wealth generation are all positively related to SCS service performance [29]. Nevertheless, most studies thus far have followed the “indicator design -scoring” paradigm, relying on static or explicit indicators and failing to consider or incorporate more involved dynamic interactive networks, collective actions or intra-organizational cooperation during the service process [30,31].

2.2. Social network analysis (SNA) and service performance

SNA conceptualizes individuals as nodes and the relationships among them as ties [32]. In terms of composition, a complicated social network encompasses transactional content, the nature of specific links, and structural characteristics [33]. Previous studies have provided comprehensive evidence as to the relationships that exist between SNA and organizational performance [34]. Individuals in a network contribute unequally to service performance due to differences in their own personal characteristics as well as in their strategic position [35]. By specifying whether an individual is an opinion leader, sensitive, or in a central position can be helpful in examining the effects that exist between inequality and performance [36]. Furthermore, a broader examination of the metrics regarding network structure, including clustering coefficients, degree distributions, average path lengths, and graph densities, can facilitate a more detailed analysis of the service outcomes and resource exchange efficiency of a particular group [37,38].

Current research on social networks and performance can be summarized as being either type-oriented or structure-oriented. Discrepancies in transaction content or established relationships can result in different network types. Zhang and Venkatesh studied online and offline communication networks, distinguishing direct and indirect networks, and suggest that employees' connections in both online and offline workplace communication networks were complimentary in their influence on their job performance [32]. When investing employee advice networks within an organization, Sykes et al. introduced the concepts of “get-advice” and “give-advice” networks based on the direction of information flow. Other scholars have focused on investigating the relationship between network structure and performance [39], such as Henttonen et al. who have suggested that denser relationships and more spatially dispersed instrumental networks are associated with enhanced work performance [40]. Meanwhile, Muller and Peres divided the social network structure into three levels, global, dynamic, and individual, each of which contains a specific set of indicators which can be used to evaluate the network structure, and used these to explore the impacts of each network level on the performance of four different organizational innovation adoption processes [41].

The advantages of utilizing SNA to assess the China's SCS service performance lies in its dynamism and comprehensiveness. Sci-tech commissioner networks manifest randomly and intricately, and contain variability in their relationships which can be described as complex, entangled, open, fluid, or dynamic [42]. As such, adopting methods which assume logical, ordered, or linear processes (e.g., in designing scoring standards) can be not only difficult but often useless [43]. Furthermore, public service social networks encompass a wide variation of individual beliefs and social capital, which are hard to measure using objective data indicators [44]. Due to its complex use of network diagramming as well as mathematical metrics, however, SNA can be used to achieve a much more nuanced understanding of the various relationship networks at play within the SCS, and their impacts on its service performance [42,45].

2.3. Information systems and technology, machine learning, and service performance

The ongoing development of the information systems and technology sector has reinforced the digitalization of service performance management. In the context of an information system where data mining techniques can be used to discover untapped patterns within data, technologies such as data warehousing, data mining and other artificial intelligence technologies can help in the evaluation of knowledge creation, as well as in data storage, dissemination, and management processes [46,47]. In other words, information systems can enhance service performance in numerous ways, such as leading to better decision-making, improving customer feedback management and allowing for greater operational efficiency [48]. Latifah et al. have suggested that information systems should be an integrated part of management control to reduce risk by identifying standards and directing organizational goals [49]. Meanwhile, increased proficiency in information system and technology use was shown to significantly improve employees' efficiency and job performance [50]. The integration of information technology has also been confirmed to promote service performance in the agricultural sector. For example, the use of information communication technology has led to increased quality of service extension and agent accountability, and helped to determine what works best in extending service networks and diagnosing problems [51]. Additionally, information communication technology use in the agricultural sector has the potential to accelerate and broaden information diffusion to farmers and reduce gender bias [52].

The emergence and adoption of ML also makes it possible to facilitate and increase communication or collaboration between sci-tech commissioners to improve performance analyses. First, ML is able to learn complicated relationships between inputs and outputs from empirical datasets, extract high-dimensional features, and process nonlinear data [53]. This autonomously taps into the relations and rules behind the data, rather than depending on artificially predefined rules between variables, making it more suitable for dealing with the complex networks of interpersonal relationships of the SCS. Underwood et al. have pointed out that ML tools have become favored over traditional statistical methods for dimension reduction, feature extraction, and trend identification, all of which can be

done without configuring the econometric model beforehand [54]. Second, ML can automatically process and analyze large-scale datasets, significantly improving data analysis capacity and fault tolerance. Zhu et al. found that ML algorithms demonstrated superior learning performance from extensive datasets, resulting in more accurate predictions and insights, especially in situations with vast and complex data [55]. More importantly, the appearance of interpretable algorithm, such as SHAP, makes it possible to explore the “black-box” mechanism of ML. One insight that SHAP is able to provide us with, in terms of the inner workings of models, is gradient boosting, which can often produce opaque outputs, allowing us to measure feature importance by assessing the impact of perturbing features on prediction errors [56]. As such, scholars are increasingly integrating ML with performance analysis. For example, Song et al. utilized the Least Squares Twin Support Vector Machine to predict business performance through financial ratios [37], while Jin et al. incorporated multiple additive regression tree, neural network, support vector machine, and linear regression modes to examine the dynamic relationship evolution between the depth and breadth of knowledge and innovation performance [57].

To sum up, despite numerous studies regarding the impacts and integration of the science and technology and agricultural sectors, few have examined the service performance of the agricultural extension system in China, especially taking a micro perspective to consider agents and interactive networks. Due to the transition to management digitalization, sci-tech commissioners have adopted a cloud-based platform that records the activities, interactions, and operation process of service agents, which now allows for the investigation of the influence of personnel interactions on the service performance of SCS in China. As such, this study established a network understanding of commissioner relationship and activities, and determining structural characteristics of the various group types. This study also utilized ML to elaborate on the determinants of the SCS performance and made use of the SHAP algorithm to

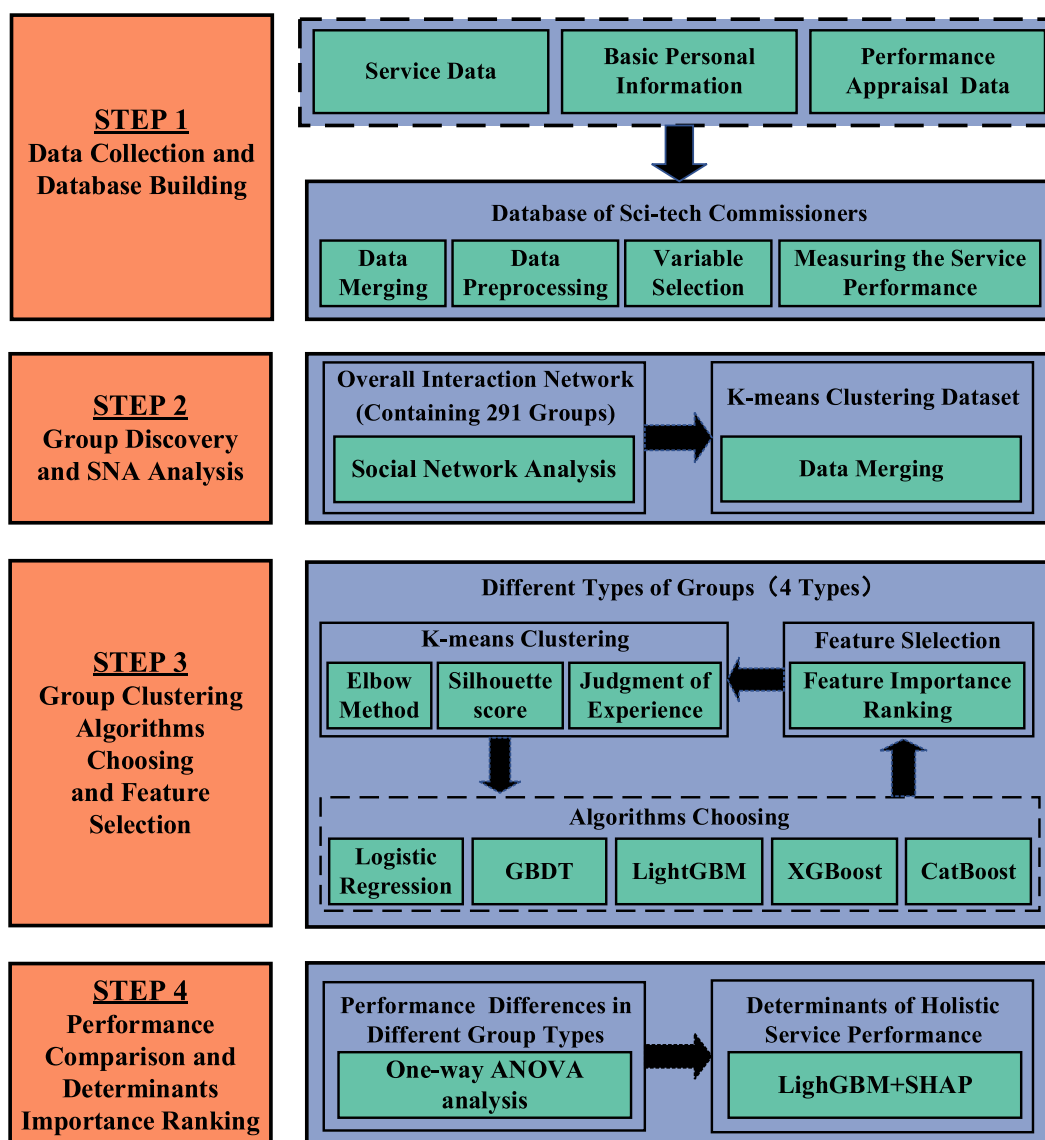


Fig. 1. Research framework.

visualize the results.

3. Methodology

3.1. Research framework

The research process was divided into four steps: 1) data collection and database building, 2) group discovery and social network analysis, 3) group clustering, algorithms choosing and feature selection, 4) performance comparison and analysis of determinants' importance. The detailed research framework is presented in Fig. 1.

Data collection and database building. i) Data cleaning. We use commissioner's phone number as a unique identifier, and merge the basic personal information, service data and performance appraisal data to establish an analytical database. ii) Data preprocessing. Cleaning missing values, duplicate values, and noise values in the sci-tech commissioner analysis database and converting categorized data to meet the requirement of the actual situation. iii) Variable selection. Screening variables in the database to retain those consistent with the study objectives and ensure the analytical accuracy. iv) Measuring the service performance. Calculating service performance scores based on the service performance scoring rules for sci-tech commissioners.

Group discovery and group clustering. i) Taking an overview of the overall interaction network. Using the sign-in data of the sci-tech commissioner database, SNA was used to illustrate the overall service interaction network and analyze the degree centrality, betweenness centrality, and closeness centrality of the nodes within it. ii) Data merging. The indicators generated from SNA were merged with those retained from the variable screening process to form clustering analysis dataset.

Group clustering, algorithms choosing and feature selection. i) Group clustering. Performing initial clustering by using the K-means algorithm, and determining the optimal cluster number by combining the elbow method, the Silhouette score method, and practical experience. ii) Algorithms choosing. Comparing traditional prediction algorithms with integrated learning algorithms to screen out the one that predicts clustering results most accurately as a K-means clustering feature selection algorithm, avoiding possible dimensional catastrophe problems in the clustering process. iii) Feature selection. Based on the screening algorithm, the importance of the features was ranked, and the final features involved in K-means clustering were screened in the context of the actual situation.

Performance comparison and analysis of determinants' importance. i) Comparing between-group performance differences. One-way ANOVA analysis was used to examine the differences in the service performances of the different groups. ii) Analysis of determinants' importance. LightGBM (Light Gradient Boosting Machine) and SHAP algorithms were used to explore the determinants affecting the overall performances of each group. Finally, discussions were had within the research team to assess the results, and confirm the importance of the various determinants according to each individual group type.

3.2. Data collection

1. Data source

The data used in this study was provided by the science and technology department of Province Y, China, and comprised the personal information of all sci-tech commissioners working in Province Y in 2022, a total of 4885 records. The data covered 28 different attributes, including gender, birthplace, ethnic group, title, years of employment, birthdate, political profile, workplace level, and service industry. We also obtained the 2022 service data of the sci-tech commissioners working in Province Y, which covered 15 attributes including service area, base name, time in service, service category, and service industry. Finally, we also obtained the 2022 merit data of the sci-tech commissioners working in Province Y in 2022, which comprised 51 attributes including performance score, issues resolved, and numbers of trainees, remote guidance, and live training sessions.

The final comprehensive dataset of the sci-tech commissioners examined in this study, once all data from the multiple sources was integrated, comprised approximately 80 unique attributes. Based on our research objectives, we selected 28 attributes for data mining and analysis. Details of the specific data variables included in this study and their descriptions are shown in Appendix 1. Altogether, the included personal information and evaluation data in this study amounted to 4885 entries, and service data included 213,402 entries.

2. Data preprocessing

We used the following three steps to preprocess the data to improve computational efficiency and prediction accuracy. First, we cleaned the noise and outliers in the data. For example, according to the age regulation of the sci-tech commissioner, those more than 70 years old or less than 20 years old do not meet the hiring requirements, and for this kind of data, we excluded them from the analysis. Secondly, the variables were transformed to meet the model requirements. For numerical variables we input them directly into the model, and for categorical variables, we converted them into dummy variables before inputting them into the model. Finally, for null and missing values in the dataset, e.g., data with null or zero days of service, we deleted them directly. In the end, after data preprocessing, 4885 commissioners' data were retained in the dataset.

3. Variable screening

In this study, we were concerned with the key factors affecting the service performance of sci-tech commissioners, so further screened the above 28 variables accordingly. First, based on previous studies, individual characteristics, such as age and gender, had a

significant impact on service performance [58,59], so such personal characteristic variables were all included in the analysis of influencing factors. Secondly, the data of commissioner services, such as the number of offline and online training, were directly related to the performance evaluation, and if including them in the analysis, it might cause the confusion of independent and dependent variables, so they were only used to support the calculation of performance scores and not in the research of impact factors. Similarly, so did the data on the performance evaluation of the commissioner service. In this way, we effectively avoided potential data analysis bias through variable screening. Ultimately, the variables used for impact factor analysis are shown in Table 1.

3.3. Measuring the service performance of sci-tech commissioners

Performance score is a key metric in rating the service quality of sci-tech commissioners. It is not only crucial for evaluating individual performance, but also for computing average group performance scores. An individual sci-tech commissioner performance score encompasses 12 evaluative dimensions, including service check-in, service base, sci-tech training sessions, live-streaming training sessions, information dissemination, remote guidance, district size, verification and evaluation of activities, service recipient satisfaction, regional conditions, number of service recipients, entrepreneurship and interest development, and service recognition. Each variable category is measured using specific scoring criteria. For example, sci-tech training is scored by number of sessions, with each session scores 2 points, plus another 0.2 points added for each trainee participant; the live-streaming training score considers viewership numbers, with fewer than 100 people adding 0.5 points, 100 to 300 people adding 1 point, 300 to 500 people adding 1.5 points, and more than 500 people adding 2 points. The final performance score for each commissioner was calculated by summing the scores of all the subdivided criteria. The specific calculation method for the individual performance scores of the sci-tech commissioners is shown in Appendix 2. The maximum score refers to the maximum points that could be obtained for that particular item. Average group performance score was determined by averaging the sum of all individual commissioner performance scores within the particular group.

3.4. Social network analysis

Gephi, a social network analysis software, was used in this study to calculate the in-degree, out-degree, betweenness centrality, and closeness centrality for each node in the sci-tech commissioner network, to indicate its importance in the network [61]. The Statistical Inference module of Gephi, which follows the Bayesian inference optimization criteria and adheres to the same convergence heuristic as the Louvain algorithm [62], is useful for classifying nodes into different groups. Therefore, this module was utilized to infer the potential groups within the commissioner network and calculate the graph density, clustering coefficient, and average path length of each group to demonstrate the closeness of the connections between nodes. Gephi was also used as it can serve as a visualization tool to map and display the social networks.

The following structural characteristics were adopted in our SNA:

Eigenvector centrality leverages the eigenvectors to assess the centrality of individual nodes, considering both the structure of the network and the centrality of neighboring nodes. A larger value indicates that the nodes, representing the SCS service agents in a specific group in our analysis, are more closely connected with others, and collaborate more frequently with each other. This was calculated using the following formula:

$$x_i = \frac{1}{\lambda} \sum_{j=1}^n a_{ij} x_j \quad (1)$$

In Eq. (1), λ is the eigenvalue corresponding to the largest node, n is the number of nodes in the graph, x_j is the eigenvector centrality of node j , and the elements a_{ij} of the adjacency matrix represent whether there is an edge connecting node i and node j . If there is an edge connecting them, the value is set to 1, otherwise, it is set to 0.

Table 1

13 selected variables and explanations.

Variables	Variable Description
Gender	Commissioner's gender: male/female
Place of Origin	Commissioner's place of origin: local/non-local
Ethnicity	Commissioner's ethnicity: Han/minority
Work Experience	Commissioner's years of work experience: range [1,50]
Age	Commissioner's age: range [20,60]
Political Affiliation	Commissioner's political affiliation: Communist Party of China/other party/non-affiliated
Professional Title	Commissioner's professional title: junior/middle/senior/senior+/other
Educational Background	Commissioner's educational background: technical secondary school/junior college/bachelor's degree/master's degree or above/other
Commissioner Type	Type of commissioner: provincial/municipal/county/other
Service Industry	Commissioner's focus industry: planting/breeding/other
Affiliated Institution	Name of the institution to which the commissioner belongs
Service Days	Number of days the commissioner served, range [1275]
Number of Scientific Achievements	Number of scientific achievements published by the commissioner in the current year: range [0,15]

Closeness centrality reflects the position and influence of the commissioners in the particular group network. The larger the group is, the easier it is for the individual service agents in the group to establish connections with other agents, giving them higher visibility in the network and higher collaborative ability. Mathematically, the closeness centrality $C(i)$ of a node i is calculated by:

$$C(i) = \frac{1}{\sum_{j \neq i} d(i, j)} \quad (2)$$

In Eq. (2), $d(i, j)$ represents the shortest path length from node i to node j .

Betweenness centrality represents the mediating role of the service agents within their particular group. A larger average betweenness centrality means that the SCS service agents are more influential within their group, and that they are able to connect to other agents more quickly. The betweenness centrality $B(i)$ of a node i is calculated as the ratio of the number of shortest paths between all node pairs that pass through that particular node as follows:

$$B(i) = \sum_{s \neq i \neq j} \frac{\sigma_{st}(i)}{\sigma_{st}} \quad (3)$$

In Eq. (3), σ_{st} represents the total number of shortest paths between nodes s and t , and $\sigma_{st}(i)$ denotes the number of those shortest paths that traverse node i .

Graph density refers to the ratio between the actual number of connections present in a network and the total number of possible connections. Essentially, it quantifies the level of connectivity among the nodes within the network.

$$D = \frac{2m}{n(n-1)} \quad (4)$$

In Eq. (4), m represents the total number of edges in the network, and $n(n-1)/2$ represents the total possible number of edges in an undirected graph.

The clustering coefficient quantifies the extent to which the neighbors of a node are also connected to one another, reflecting the closeness of the connection and cooperation among the group members. The larger the indicator, the closer the collaboration among the members of the group.

$$CC_i = \frac{2E_i}{k_i(k_i - 1)} \quad (5)$$

In Eq. (5), E_i is the number of edges between the neighbors of node i , and k_i denotes the number of neighbors of node i .

Average degree refers to the mean number of connections each node has in a network. The average degree provides insight into the overall connectivity of the nodes in the network. The larger the value, the higher the level of group collaboration. Average degree k is calculated by:

$$k = \frac{1}{n} \sum_{i=1}^n d_i \quad (6)$$

In Eq. (6), n is the number of nodes and d_i is the degree of each node i .

3.5. Clustering and predictive algorithms

In order to classify the types of sci-tech commissioner groups, the K-means clustering algorithm, which is efficient in performance and widely used in group classification [63], was chosen for this study. Since the K-means clustering process relied on the Euclidean distance between the data points and the clustering centers, the fact that the variables in the algorithm were equally weighted leads to the possibility of dimensional catastrophe when the number of variables was too large. That is, when the difference in the distances between the data points reduces, the effect of the clustering could also decline, making it more difficult to interpret the results. In this study, there were as many as 38 features involved in clustering (Appendix 3), and it was obviously not desirable to use all the features directly for clustering. To address this challenge, the research group first compared the prediction effect of traditional prediction algorithms (logistic regression) with tree-based ensemble learning algorithms (GBDT, XGBoost, CatBoost, LightGBM) on the results of K-means clustering to select the optimal model. Then, with the help of feature importance assessment method, the feature set having the most influence on the clustering results was selected from many features, and the optimized clustering analysis was carried out in this way to obtain more accurate and easy-to-understand clustering results. The specific clustering and prediction algorithms are explained below.

1. K-means clustering

After detecting 291 sci-tech commissioners groups using the Gephi community discovery algorithm, the K-means clustering algorithm was applied from the SKlearn ML library to summarize the different group types. This algorithm is commonly used in data mining to find homogeneous data and distinguish heterogeneous data. It first divides the data into K clusters, then calculates the Euclidean distance from each sample to each cluster, and assigns it to the nearest cluster. The process can iterate until the samples

within each cluster reaching maximum similarity, while the differences between clusters achieve minimum similarity.

The K-means clustering algorithm usually requires researchers to determine the number of clusters manually. K is generally determined using the elbow method and Silhouette score. The elbow method means that as the value of K increases, the square sum of clustering error gradually decreases. When a significant inflection point exists in the magnitude of decrease in the square sum of clustering error, the value of K corresponding to the inflection point is the optimal number of clusters. The Silhouette score is a metric used to calculate the goodness of a clustering algorithm. It measures how close each point in one cluster is to the points in the neighboring clusters. Its values range from -1 to 1 , where a high value indicates that the object is well matched to its own cluster and poorly matched to neighboring clusters.

2. Logistic Regression

Logistic regression is commonly used for binary classification problems, but it can be extended to multi-categorization problems [64]. Two common methods for handling multi-class problems are One-vs-Rest (OvR) and One-vs-One (OvO), where One-vs-Rest for each category is modeled by pitting that category against all other categories. One-vs-One builds a model for each pair of category combinations. This method generates a large number of models when there are many categories (e.g., having N categories will require $N(N-1)/2$ models) [65]. Logistic regression is often extended to multi-categorical problems using the OvR strategy because of the simplicity of implementation and the relatively low computational effort of this method. The logistic regression mathematical expression is as follows:

$$S(z) = P\left(\frac{1}{1 + e^{-z}}\right) \quad (7)$$

In Eq. (7), S is the output (probability estimate) between 0 and 1, z is the input to the function, and e is the base of the natural logarithm.

3. Gradient Boosting Decision Tree (GBDT)

The GBDT algorithm is a tree-based ensemble algorithm proposed by Firedman, which has been widely used in regression. Unlike the Random Forest algorithm RF or the traditional single-tree model, the GBDT algorithm reduces bias and produces a combination of trees, while the RF algorithm reduces variance. Details about the GBDT algorithm can be found in Firedman's book [66].

4. Extreme Gradient Boosting (XGBoost)

XGBoost is an efficient algorithm for implementing the gradient boosting framework proposed by Chen and Guestrin, which is based on the integrated learning technique of decision trees and is widely used in classification, regression and sorting tasks. The algorithm uses a new regularization technique to control overfitting, which is the main difference between XGBoost and other gradient boosting algorithms. The mathematical expression of XGBoost is as follows:

$$L(f) = \sum_{i=1}^n L(\hat{y}_i, y_i) + \sum_{m=1}^M \Omega(\delta_m) \quad (8)$$

In Eq. (8), $\Omega(\delta)$ is $\alpha|\delta| + 0.5\beta\|w\|^2$, $|\delta|$ is the number of branches, w is the value of each leaf, and Ω is the regularization function. XGBoost uses a new gain function with the mathematical expression:

$$P(X_j = x_j | Y = C_k) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(x_j - \mu_k)^2}{2\sigma_k^2}\right) \quad (9)$$

$$G_j = \sum_{i \in I_j} \mathcal{J}_i H_j = \sum_{i \in I_j} h_i \quad (10)$$

$$Gain = \frac{1}{2} \left[\frac{G_L^2}{H_L + \beta} + \frac{G_R^2}{H_R + \beta} - \frac{(G_R + G_L)^2}{H_R + H_L + \beta} \right] - \alpha \quad (11)$$

In Eqs. (9)–(11), $\mathcal{J}_i$ is for $\partial_{y_i} L(\hat{y}_i, y_i)$ and h_i for $\partial_{y_i}^2 L(\hat{y}_i, y_i)$. G is the fraction of right child nodes, H is the fraction of left child nodes, and $Gain$ is the fraction in the case of no new child nodes [67].

5. Categorical Boosting (CatBoost)

CatBoost is a newly developed GBDT algorithm, which has the following improvements over the traditional GBDT algorithm: (1) When dealing with categorization features, if the deposit arrangement is used:

$$x_{\sigma,p,k} = \frac{\sum_{j=1}^{p-1} [x_{\sigma,p,k} = x_{\sigma,p,k}] \bullet Y_{\sigma j} + \beta \bullet P}{\sum_{j=1}^{p-1} [x_{\sigma,p,k} = x_{\sigma,p,k}] + \beta} \quad (12)$$

In Eq. (12), β and P are the weights of the prior and the prior value, respectively; (2) CatBoost employs a technique called ordered augmentation for solving the gradient bias problem; (3) the algorithm uses a forgetting tree as the basic predictor; and (4) in order to avoid overfitting, the algorithm employs a new model in the selection of the tree structure to compute the values of the leaf nodes [68].

6. Light Gradient Boosting Machine (LightGBM)

LightGBM implements the GBDT algorithm framework for efficient parallel training and was developed to address the issue of time consumption with large, high-dimensional data samples [69]. The algorithm achieves these goals by reducing the number of data instances and features. It employs two new techniques to achieve this. The first one is Gradient-based One-Side Sampling (GOSS), which strikes a good balance between reducing data instances and maintaining the accuracy of the decision tree. The second is Exclusive Feature Binding (EFB), which can effectively achieve the purpose of reducing the number of features.

3.6. The SHAP (SHapley additive explanation) algorithm

While a ML model based on an integrated algorithm, such as LightGBM will perform better, its increased complexity diminishes its interpretability, rendering the ML model somewhat of a “black box” model. However, interpretability and comprehensibility are

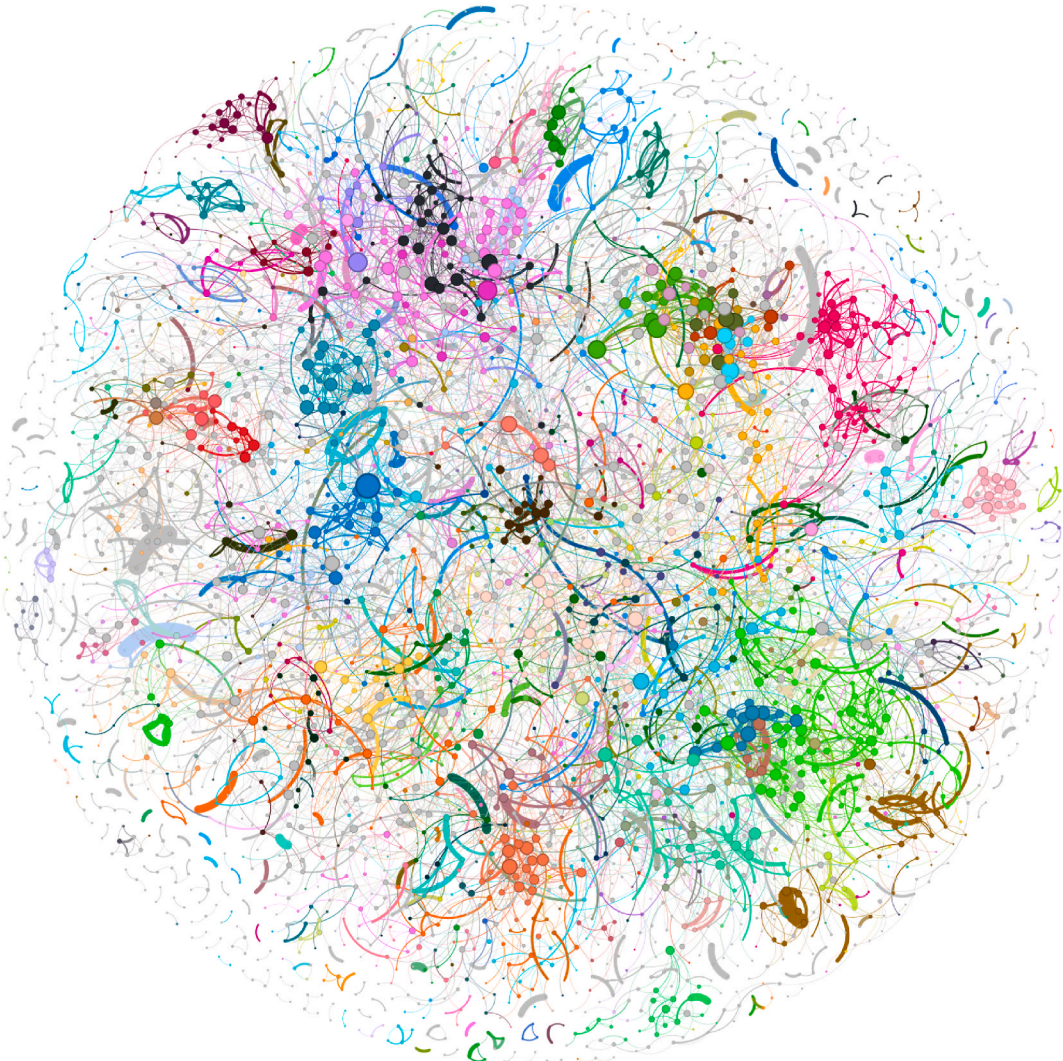


Fig. 2. The overall service network of province y sci-tech commissioners.

critical to assessing the service performance of sci-tech commissioners, particularly for use in the presentation of nuanced prediction model results to policy stakeholders. To address these challenges of interpretability the SHAP algorithm was introduced to further clarify the ML results. The SHAP algorithm is a unified framework for interpreting predictions by computing the contribution of each feature individually to the overall prediction [70]. SHAP constructs an additive explanatory model where all features are considered “contributors” [60]. The mathematical expression for SHAP is as follows:

$$\varphi_i(f, x) = \sum_{S \subseteq \{1, 2, \dots, p\} \setminus \{i\}} \frac{|S|!(p - |S| - 1)!}{p!} [f(x_S) - f(x_{S \cup \{i\}})] \quad (13)$$

In Eq. (13), $\varphi_i(f, x)$ represents the SHAP value of feature i for model f on sample x . f denotes the model. x represents the input sample. p denotes the number of features. S denotes a subset of features. x_S represents a sample with only the features from subset S .

SHAP is consistent in both local and global explanations and can illustrate the positive/negative relationship between predictors and target variables. For local interpretability, each feature is endowed with a set of Shapley values which can be used to explain the contribution of each feature to the prediction. For global interpretability, a feature’s importance can be determined by averaging the Shapley values corresponding to that feature across all samples. Thus, the advantage of the SHAP algorithm is that it can reflect the influence of each feature as well as the positivity or negativity of its influence, revealing how that particular feature affects the prediction results.

4. Results

4.1. The overall network analysis of the sci-tech commissioners in province Y

The results of our analysis of the sci-tech commissioner service data were used to generate a global network image of the SCS depicting each commissioner and their interactions as nodes and edges, respectively (see Fig. 2), which clearly presents both the group-based and individual-based services provided by the commissioners. Commissioners who conducted group-style services are located at the center of the network (dots of the same color belong to the same group), while commissioners who acted individually are scattered throughout the network.

Further analysis was conducted on the resulting network, with detailed results shown in Table 2. The average weighted degree of the service network was 13.955, suggesting that, on average, each commissioner maintained interactions with at least 13 other participants, and that communication and cooperation between commissioners is very active. Moreover, the average clustering coefficient of the network was 0.398, indicative of tight localized connectivity within the network. A cohesive clique was identified among the sci-tech commissioners, found mainly in the groups which belonged to the same science and technology service unit. The average path length was 7.530, implying that, under worst-case conditions, 7 to 8 intermediary nodes are needed to establish a connection between any two sci-tech commissioners.

Key nodes play a vital role in group-based science and technology services. These are the leaders and planners of group-based services, who exert more influence than other members in the network. In this study, nodes with “degree” of more than two were considered to be key nodes. Our analysis results indicated that 2394 (50.474 %) of the commissioners met this criterion. This implies that approximately half of all the commissioners kept connections with more than two peers, facilitating the efficient flow of information and resources throughout the network.

Finally, the Statistical Inference module of Gephi was used to group all network nodes according to the similarity of their connection patterns within the network structure. This process identified 291 naturally formed communities, providing critical support for subsequent group-type clustering and performance analysis.

4.2. The group types of province Y the sci-tech commissioners

To identify the features of the various Province Y sci-tech commissioner groups, including their structural characteristics and operational dynamics, K-means cluster analysis was conducted on the 291 commissioner groups identified by the Gephi community discovery algorithm in (see Section 4.1). Meanwhile, we constructed a K-means clustering feature set based on the analytical framework in Section 3. This feature set not only included the metrics screened in Section 3.2, but also incorporated the metrics obtained from Gephi’s social network analysis (see Appendix 3 for details). Then, we input them all into the K-means clustering algorithm and determined the optimal number of clusters by using the Elbow Method (see Fig. 3-(a)), Silhouette scores (see Fig. 3-(b)),

Table 2
Parameters of the sci-tech commissioner social network.

Parameters	Value
Nodes	4743
Edges	8969
Average weighted degree	13.955
Average clustering coefficient	0.398
Average path length	7.530
Ratio of key nodes to all nodes	50.474 %

and background experience. Further, by comparing the effects of each prediction algorithm in Section 3.5 under the default parameter settings, we found that the LightGBM model most accurately predicted the results of K-means clustering (see Table 3). After that, we used LightGBM as the core algorithm, and adopted feature importance analysis method to filter out the 16 features that had the greatest influence and the strongest explanatory power on the prediction effect of the model (see Fig. 3-(c)). Based on these 16 characteristics, we re-clustered the sci-tech commissioner groups and finally categorized them into four different types, plus one more comprising those who had not been involved in any cooperation with others during the data collection period, meaning five specific types of groups (see Table 4).

Cluster 1 is made up of small groups of young commissioners exhibiting close cooperation with others. This group type had a smaller average number of members (2.320), most of whom come from the same organization. Their average age was lower (41.433), as was their seniority in their position compared to that of other clusters, with an average of 11.817 working years. Their average service duration was not long, at only 39.343 days. Cluster 1's average eigenvector centrality of 0.017 indicates that the nodes within the cluster do not occupy a central position in the overall network. An average closeness centrality of 0.913 and an average graph

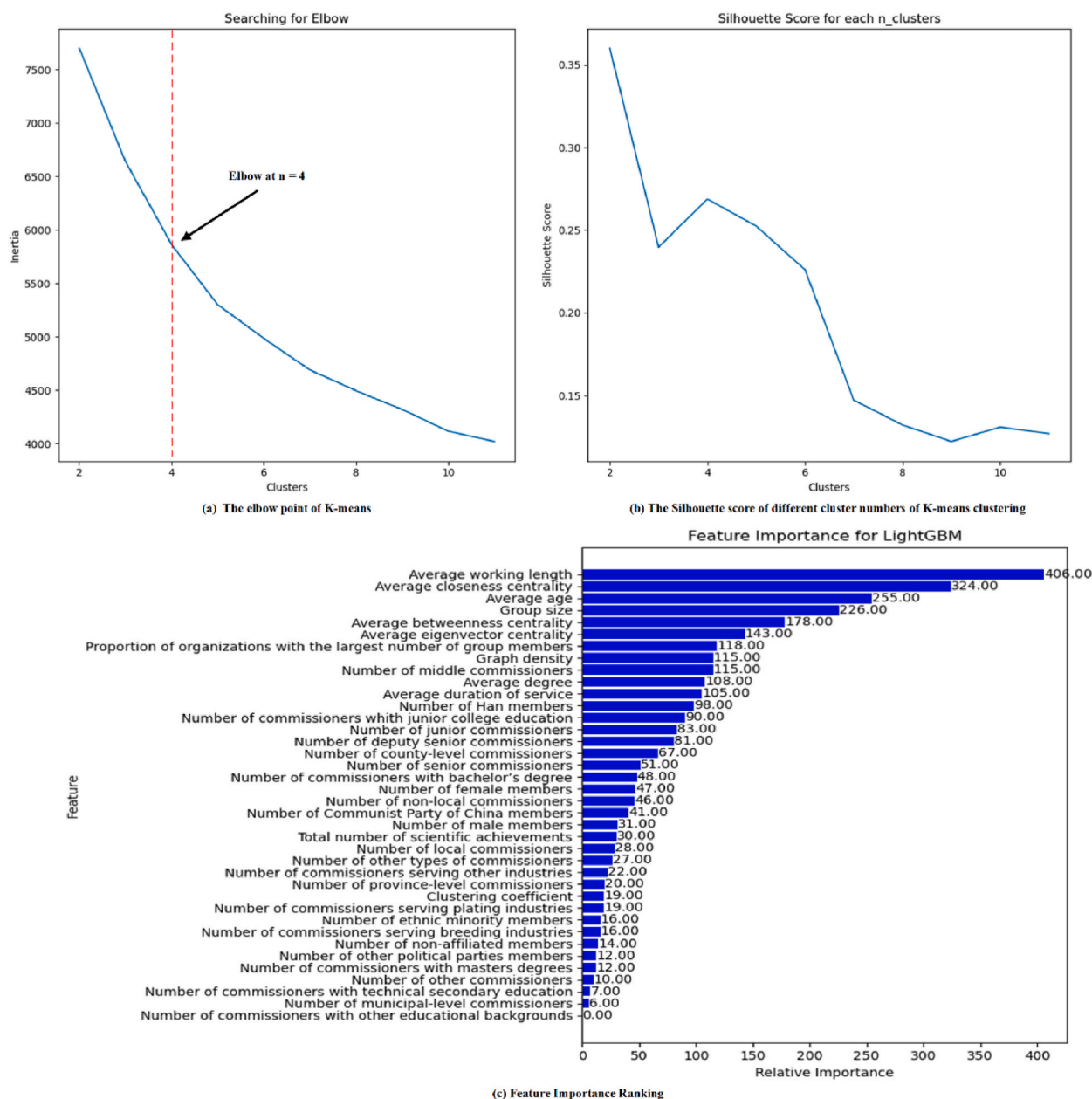


Fig. 3. Determination of optimal number of clusters and feature selection. (a), (b) and (c) respectively represent the elbow point, the Silhouette point and feature importance ranking.

Table 3

Comparison of prediction algorithms.

Model	Accuracy	ROC_AUC	Precision	Recall	F1 Score
Logistic Regression	0.924	0.993	0.921	0.911	0.916
GBDT	0.945	0.996	0.932	0.937	0.934
XGBoost	0.969	0.997	0.963	0.939	0.951
CatBoost	0.967	0.996	0.963	0.943	0.953
LightGBM	0.971	0.998	0.974	0.942	0.958

Table 4

K-means clustering results of province y sci-tech commissioners.

Indicators	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Group of those acting individually
Average group size	2.320	25.080	2.650	1.000	1027.000
Mean of proportion of organizations with the largest number of group members	0.765	0.386	0.768	1.000	0.010
Mean of senior commissioners	0.120	2.140	0.070	0.000	35.000
Mean of deputy senior commissioners	0.300	9.170	0.680	0.500	237.000
Mean of middle commissioners	0.990	11.070	1.660	0.500	490.000
Mean of junior commissioners	0.610	1.980	0.150	0.000	179.000
Average age	41.433	45.137	49.736	39.375	46.252
Average working length	11.817	17.776	24.232	11.625	17.130
Average duration of service	39.343	48.013	46.918	73.375	38.900
Average eigenvector centrality	0.017	0.031	0.000	0.727	0.000
Average closeness centrality	0.913	0.159	0.901	0.171	0.000
Average betweenness centrality	644.342	9865.809	0.227	76959.404	/
Average graph density	0.916	0.234	0.861	/	/
Average clustering coefficient	0.371	0.519	0.300	/	/
Average degree	1.089	2.782	1.159	/	/
Average performance score	115.739	131.850	129.277	153.138	113.670
Group sums	76	133	74	8	1

density of 0.916 suggests that the members within Cluster 1 are closely connected, however, the group's average performance score is lower, at only 115.739 points. See Fig. 4-(a) for details.

Cluster 2 was made up of larger groups of young and middle-aged commissioners. This group type consists of larger groups, averaging 25.080 individuals, who are demographically diverse and do not necessarily come from one single organization. On average, they hold higher-ranking titles and their average age is 45.137. They're relatively experienced in their positions, with an average working length of 17.776 years. The average duration of service throughout the year was 48.013 days. The average eigenvector centrality of the members in this cluster is 0.031, indicating that most nodes in the cluster are not in the most central nor influential positions within the network. Due to the large number of nodes in the cluster, the average distance between nodes in the network is longer, resulting in lower values for average closeness centrality and average graph density, which are 0.159 and 0.234, respectively. The average betweenness centrality of the nodes in Cluster 2 is 9865.809, meaning that some nodes play a significant role in connecting various parts of the network. The average clustering coefficient is 0.519, higher than the overall network's average clustering coefficient, suggesting that nodes in Cluster 2 tend to form tightly-knit groups. The average group performance score was 131.850 points, highlighting high efficiency in their scientific and technological services. See Fig. 4-(b) for details.

Cluster 3 comprises small groups of middle-aged and senior commissioners who exhibit close cooperation. This cluster is composed of middle-sized groups, averaging 2.650 persons, who typically come from the same institution. Field investigations later revealed that the reason for this group size is primarily due to constraints on the number of passengers that transportation vehicles can carry, as well as a result of commissioners' efforts to save on travel costs (traveling together reduces commissioners' travel expenses, and most vehicles used for work trips are five-seaters). Members of this cluster generally came from the same institution, with an older average age compared to other clusters (49.736 years). These commissioners also possess considerable professional experience, averaging 24.232 years of service. Members rarely acted as intermediaries or bridges for the flow of information or resources within the overall commissioner network, with an average betweenness centrality of only 0.227. The connections among members are close, however with an average graph density of 0.861 and an average clustering coefficient of 0.300, indicating a low degree of interconnectedness between neighboring nodes. Given these metrics, the commissioners in this cluster typically have a core leader responsible for initiating and organizing group activities. The average performance score of the group was relatively high at 129.277, in Fig. 4-(c).

Cluster 4 was made up of isolated groups of highly influential commissioners. This type was singled out due to their appearance in multiple group types simultaneously, making it difficult to categorize them into any one specific group. Commissioners in this cluster are relatively young, with an average age of 39.375 years, and have relatively low job seniority, with average working length of 11.625 years. However, they did have the largest average number of service days of all the clusters (73.375 days). They have an average eigenvector centrality of 0.727 and an average betweenness centrality of 76,959.404, indicating that nodes in Cluster 4 exert a high level of individual influence among the commissioners and occupy significant positions within the overall network, having both direct

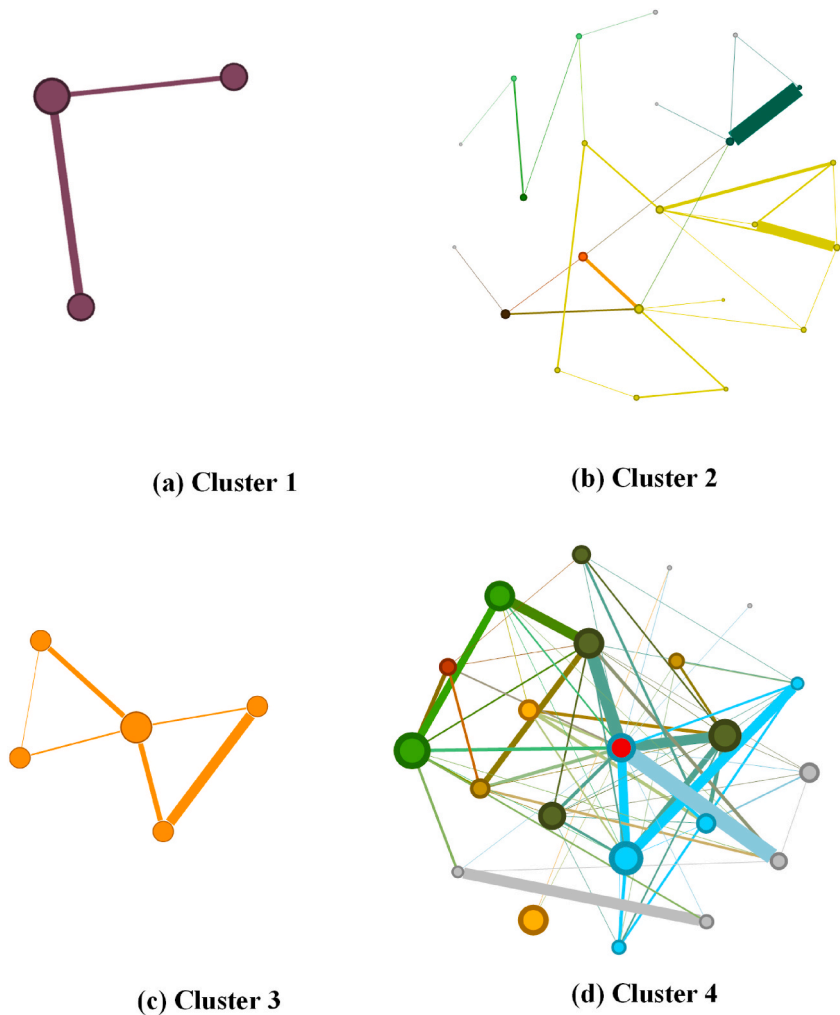


Fig. 4. Network characteristic graphs for clusters 1 to 4. (a), (b), (c) and (d) depict network interactions of each cluster.

or indirect connections with many key nodes. Correspondingly, their average performance score is exceptionally high, at 153.138, causing them to stand out in contrast to the middle-aged and young commissioners. The network characteristic graph Cluster 4 with a proximity node depth of 1 is shown in Fig. 4-(d).

The group of commissioners acting alone was not included as a separate category in either the social network analysis or the clustering analysis. Descriptive statistics indicate that the commissioners in this group are middle-aged, with an average age of 46.252 years, and have attained reasonable seniority in their positions, with the average working length of 17.130 years. They are particularly skilled to be able to conduct their roles alone. however, their average service duration is lowest, at only 38.9 days, and due to their acting individually, they scored worse in their service performance, averaging only 113.670 points.

4.3. Analysis of the performance differences between the sci-tech commissioner groups

Based on the clustering analysis results presented in Section 4.2, we observed differences in the average performance levels of the various types of commissioner groups and the individually acting commissioners, a one-way analysis of variance (ANOVA) was employed to validate the statistical significance of these differences.

First, we conducted a test for homogeneity of variances in the data, as shown in Table 5 ($P < 0.05$). The statistical results were

Table 5
Variance homogeneity test of the different group types and of commissioners acting alone.

Levene Statistic	df 1	df 2	Sig.
5.046	4	4738	0.000

significant, indicating that the samples did not exhibit variances homogeneity. Therefore, we employed Dunnett's T3 test to analyze the samples across different groups.

As Table 6 shows, there were significant differences in the various groups' service performance scores. No significant difference was seen between the Cluster 1 service performance scores and those of the individually-acting commissioners. However, significant differences did exist between the Cluster 1 service performance scores and those of Clusters 2, 3, and 4. Specifically, the average performance score of Cluster 1 was significantly lower than those of Clusters 2, 3, and 4. No significant difference was seen in the level of service performance scores between Clusters 2 and 3, however a significant difference was seen between the scores of those groups and that of Cluster 4, in that the average performance scores of Clusters 2 and 3, individually, were both lower than the average performance score of Cluster 4. The average service performance score of Cluster 4 was the highest of all the clusters.

4.4. Analysis of the factors influencing differences in group performance

The analysis results presented in Section 4.3 demonstrated that sci-tech commissioners groups across different clusters manifest distinct variations in their average technological service performance, and that concurrently, a phenomenon of performance convergence can also be seen. To clarify the significant factors impacting the average performances of these commissioner groups, a LightGBM-based multi-classification model was adopted. Then, SHAP value analysis was also utilized to identify and highlight the key determinants which substantially influence the classification outcomes.

During this process, 70 % of the commissioner group samples were classified as the training set, and the remaining 30 % were used as the test set. The model performed well under default parameters (see Table 7), and generally excels in classification, with an overall accuracy as high as 0.943. It is worth noting that, for the Cluster 4 classification, accuracy reached 100 %.

After thorough evaluation of the model's classification effectiveness, the research team employed the SHAP value analysis method to assess the importance of input features in the LightGBM model. Fig. 5 presents a bar chart showing features listed by their importance, based on the resulting SHAP values, with the vertical axis listing each feature item in order of importance (from high to low), and the horizontal axis indicating the average impact of each feature on the model's predictions. Each colored bar represents the average SHAP value of the specific feature across all data points within a particular cluster (i.e., Cluster 1, Cluster 2, Cluster 3, Cluster 4). The length of the bar signifies the average magnitude of that feature's impact on model predictions, with longer bars indicating a greater influence on the model's output. The charts show that average tenure, group size, and average closeness centrality are the three most critical features in distinguishing between clusters. It should be noted that the importance ranking of features varies across the different clusters. For groups in Clusters 1 and 3, features such as average tenure, average age, and number of commissioners with junior titles constitute key criteria for differentiating these two group types. For Cluster 2 groups, size of group, average closeness centrality, and proportion of the largest institution serve as key elements for identifying such groups. Meanwhile, graph density and average eigenvector centrality are the main characteristics to identify groups in Cluster 4.

We also harnessed the SHAP characteristic importance ranking chart and summary diagram to further clarify the contribution of each feature item to the commissioner group classification. In Fig. 6A, the horizontal axis represents the importance for each feature of Cluster1, and the vertical axis ranks all features in descending order of importance. In Fig. 6B, each colored point corresponds to a specific sample, and each row represents a feature in the model. The abscissa denotes the SHAP value of the feature item, which can be

Table 6

Dunnett's T3 post-hoc test of different commissioner group types.

Gorup types	Group Types	Mean difference	Standard Error	Sig.
Cluster 1: Small groups of young commissioners with close cooperation	Cluster 2*	-16.112	3.746	0.000
	Cluster 3*	-13.538	4.254	0.001
	Cluster 4*	-37.399	9.682	0.000
	Commissioners acting alone	2.068	3.097	0.504
	*			
Cluster 2: Larger groups of young and middle-aged commissioners	Cluster 1*	16.112	3.746	0.000
	Cluster 3	2.574	3.778	0.496
	Cluster 4*	-21.287	9.482	0.025
	Commissioners acting alone	18.180	2.400	0.000
	*			
Cluster 3: Small groups of middle-aged and older commissioners with close cooperation	Cluster 1*	13.538	4.254	0.001
	Cluster 2	-2.574	3.778	0.496
	Cluster 4*	-23.861	9.694	0.014
	Commissioners acting alone	15.606	3.135	0.000
	*			
Cluster 4: Isolated points of highly influential commissioners	Cluster 1*	37.399	9.682	0.000
	Cluster 2*	21.287	9.482	0.025
	Cluster 3*	23.861	9.694	0.014
	Commissioners acting alone	39.467	9.245	0.000
	*			
Commissioners acting alone	Cluster 1	-2.068	3.097	0.504
	Cluster 2*	-18.180	2.400	0.000
	Cluster 3*	-15.606	3.135	0.000
	Cluster 4*	-39.467	9.245	0.000

Table 7
The LightGBM prediction model classification performance.

Group Types	Accuracy	Recall Ratio	F1 Score
Cluster 1	0.951	0.873	0.910
Cluster 2	0.932	1.000	0.965
Cluster 3	0.957	0.912	0.934
Cluster 4	1.000	1.000	1.000

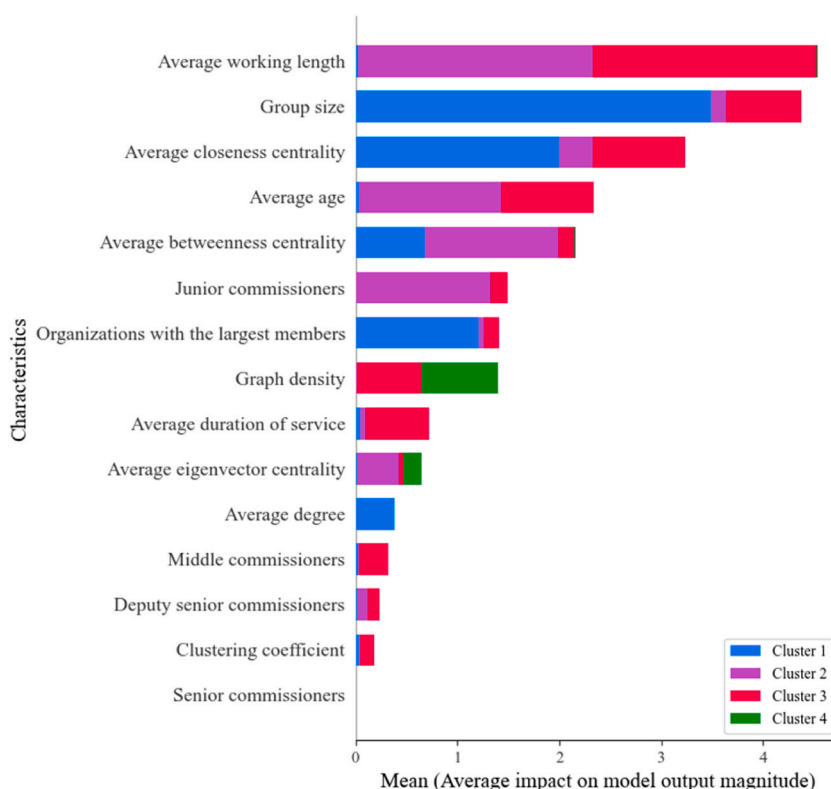


Fig. 5. LightGBM prediction model results of characteristic importance analysis.

used to reflect its influence on the prediction results for a given sample. The line where SHAP takes a value of zero is the center dividing line, and sample points to the right of the center line have SHAP values greater than zero, indicating favorable effects on the prediction outcome. And sample points that are to the left of the center line have SHAP values less than 0, indicating a negative contribution to the prediction. In this study, a positive effect indicated that a sample was more likely to be assigned to a specific group type, whereas a negative effect indicated a lower likelihood of being classified into a certain group. The point's color represents the eigenvalue level, with red indicating higher eigenvalues and blue lower ones. By observing the distribution of the colored points, we can also gain insights into how different features affect group classifications.

Fig. 6-(a) shows the main features that affected samples being classified into Cluster 1. From top to bottom, average working length contributed the most to the model prediction, which is followed by average closeness centrality, average age, group size, graph density, and average duration of service, and so on. In the feature importance ranking plot, the features, average degree and senior commissioners, are at the bottom, and their corresponding average SHAP values are close to zero. This indicates that these two features have a minimal effect on the attribution of the sample to Cluster 1, and have almost no significant impact on the model's predictions. In Fig. 6-(b), the average working length is represented by SHAP values that ranged from -4 to 4 , indicating the variation in its impact on the model output. Samples with shorter working lengths (i.e., blue dots) are concentrated on the right side of the horizontal axis, signifying a negative correlation between working length and classification in Cluster 1. Basically, the smaller the commissioners' average working lengths, the more likely they are to be classified into Cluster 1. Characteristics sharing similar negative correlations to working length include average age, group size, average service duration, middle commissioners, average betweenness centrality, organizations with largest members, clustering coefficient, and deputy senior commissioners. Lower values for these features indicated a higher likelihood of being classified into Cluster 1. Meanwhile, the higher the average closeness centrality, group graph density and junior commissioners, the more likely the samples are to be categorized into Cluster 1, demonstrating the close cooperation among its

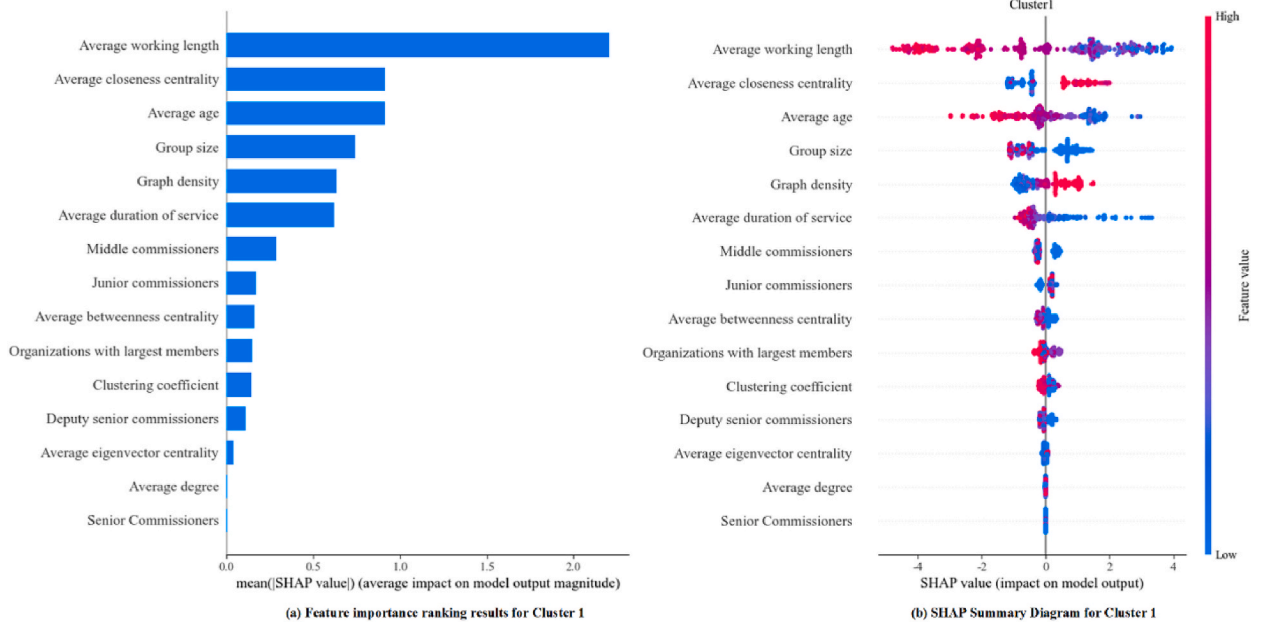


Fig. 6. Cluster 1 interpretation and analysis of SHAP. (a) is feature importance ranking results for Cluster 1 and (b) shows the SHAP summary.

members. Additionally, the relatively lower professional titles of members in Cluster 1 correspond to the previously observed lower seniority and age data within this group. This phenomenon might be one of the reasons why this group performs the least of all types.

Fig. 7-(a) illustrates the primary group features influencing the classification of samples into Cluster 2, including number of individuals, average closeness centrality, proportion of the largest institution, average betweenness centrality, and the average degree. In Fig. 7-(b), number of individuals in a group, average betweenness centrality of commissioners, and average degree of the group have a positive relationship with the samples being assigned to Cluster 2, suggesting that larger teams with closer relationships are more likely to be classified into Cluster 2. Conversely, average closeness centrality and the proportion of the largest institution both have a negative correlation with samples being classified into Cluster 2, indicating that samples are more likely to be assigned to Cluster 2 when the majority of the commissioners come from different institutions. In summary, commissioners in Cluster 2 are mostly from different organizations, and their links are relatively loose, with the groups usually connected by key nodes that serve as bridges.

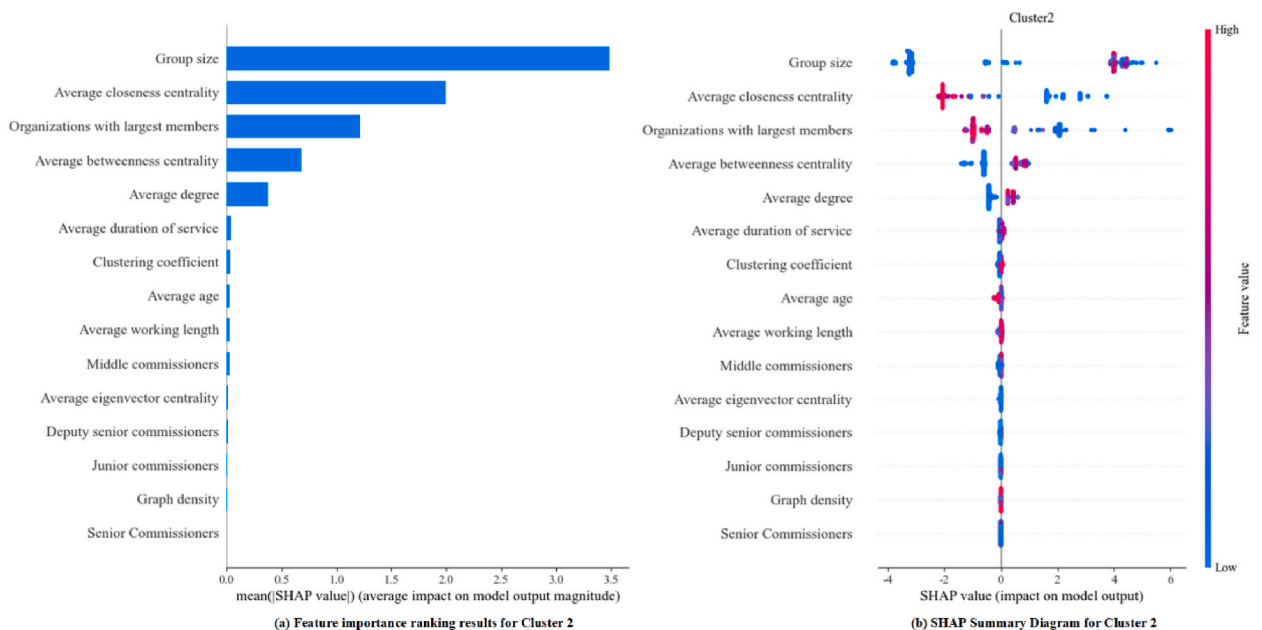


Fig. 7. Cluster 2 interpretation and analysis of SHAP. (a) is feature importance ranking results for Cluster 2 and (b) shows the SHAP summary.

According to the actual investigation, these commissioners usually belong to regional agricultural technology associations, which organize regular exchanges and training, and convene their members to provide professional and technical courses for farmers. As members in Cluster 2 regularly participate in various activities organized by the associations to help farmers, this also indirectly increases their service performance, leading to its second place among all types in terms of performance.

Fig. 8-(a) presents the primary features leading to the classification of samples into Cluster 3. The main features influencing this classification are average tenure, average age, number of junior titles, average betweenness centrality, eigenvector centrality, average closeness centrality, group size, deputy senior commissioners, average duration of service and organizations with largest members. In Fig. 8-(b), average working length, average age, average closeness centrality, average duration of service and organizations with largest members have a positive relationship with samples being assigned to Cluster 3. In contrast, number of junior titles, average betweenness centrality, and eigenvector centrality, group size, and deputy senior commissioners all have a negative relationship with samples being classified into Cluster 3. This indicates that the sci-tech commissioner groups in Cluster 3 mainly come from the same organization, and are mainly small-scale teams. The relatively small number of members makes it easier to form cohesion within the group. At the same time, the rich experience and good communication of the members ensure a long time and high-quality service output. In addition, the small proportion of less senior members and the relatively low average betweenness centrality of this cluster further confirm the close collaboration of this cluster, and the service performance of this cluster is also better than that of all other clusters.

Fig. 9-(a) shows the primary features associated with the classification of samples into Cluster 4, with the key features being graph density, average eigenvector centrality and average betweenness centrality. In Fig. 9-(b), graph density has a negative relationship with samples being assigned to Cluster 4, while average eigenvector centrality and average betweenness centrality have a positive relationship with samples being classified into Cluster 4. This implies that the more important a commissioner is within the overall network, the more connections they have with nodes in the network, and the stronger their influence, the more likely they are to be classified into Cluster 4. The “star” sci-tech commissioners in this category have the highest performance across all categories.

5. Discussions

The service performance of sci-tech commissioners plays a crucial role in the dissemination of agricultural information, enhancing production levels, and developing rural areas. Existing research on the service performance of rural sci-tech commissioners has focused mainly on the interactions between commissioners and farmers, the level of agricultural technology training, the adoption rate of technology, and investments in the agricultural industry [71,72]. These studies have often concentrated more on the behavior of individual commissioners and the support of government policies, while neglecting the impact of service models and organizational structures on service performance. In light of this, the present study conducted an in-depth analysis of the organizational forms of China's rural sci-tech commissioners, reveals different types of commissioner groups, evaluates performance differences between these groups, and discusses the key factors leading to these performance differences.

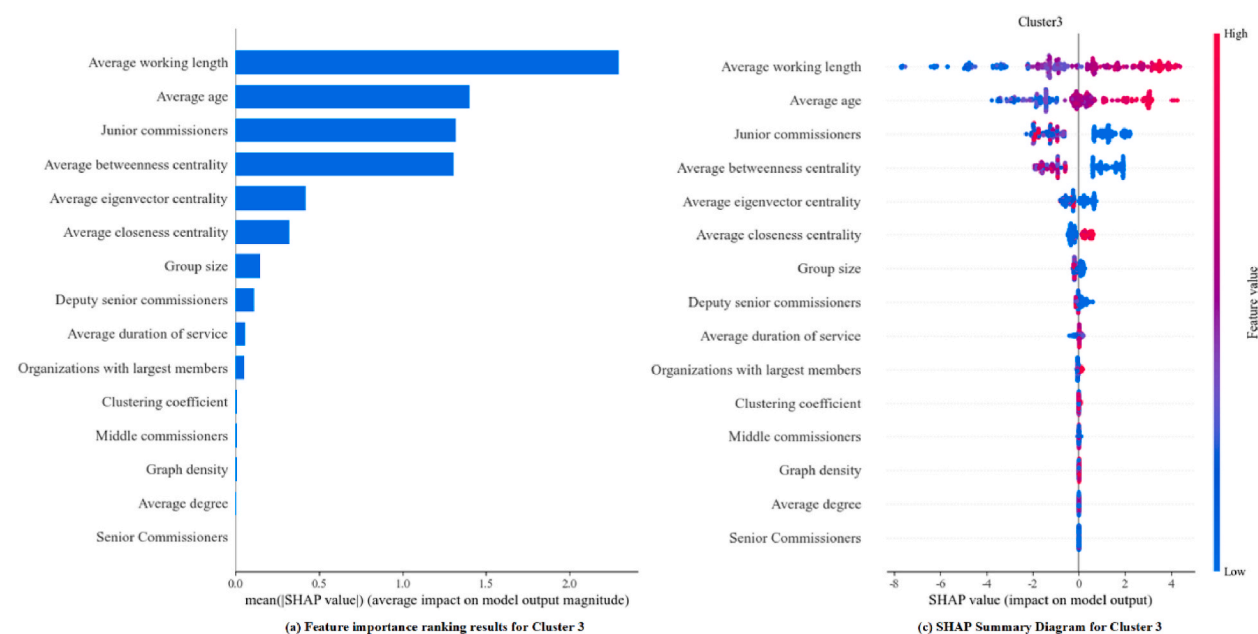


Fig. 8. Cluster 3 interpretation and analysis of SHAP. (a) is feature importance ranking results for Cluster 3 and (b) shows the SHAP summary.

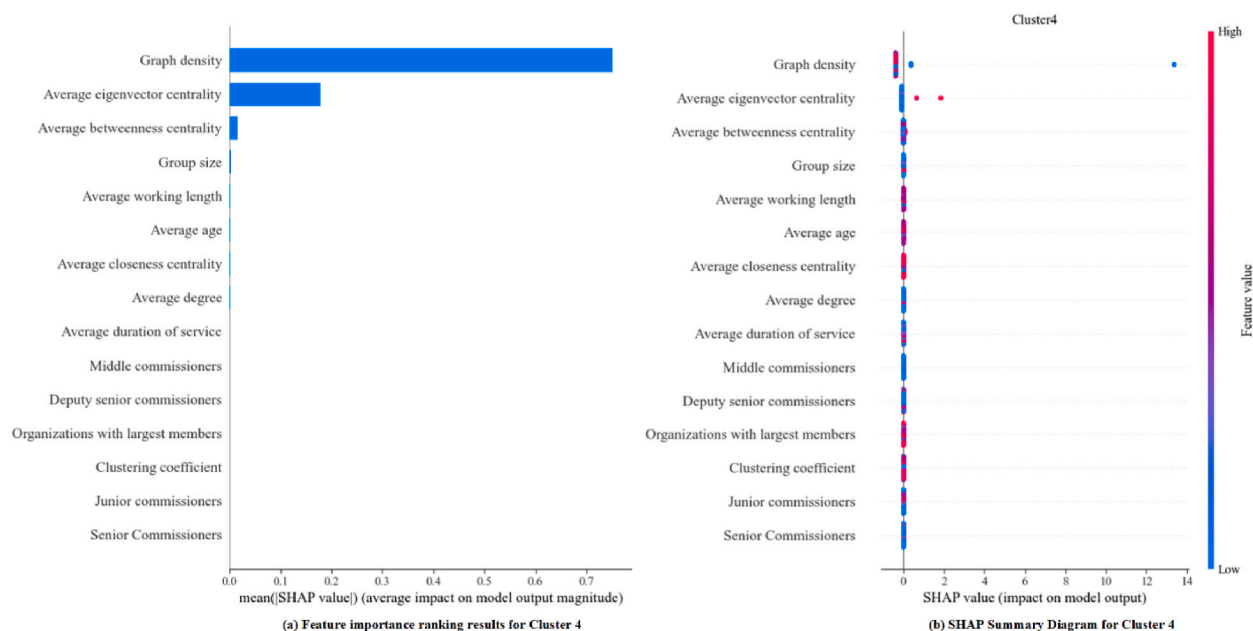


Fig. 9. Cluster 4 interpretation and analysis of SHAP. (a) is feature importance ranking results for Cluster 4 and (b) shows the SHAP summary.

5.1. Theoretical implications

Using service data provided by the Science and Technology Department of Y Province, this study applied social network analysis methods to map out the complete service network of sci-tech commissioners in Y Province. The results show that the commissioners' services are carried out primarily in groups, with 73.247 % of commissioners participating in at least one group service activity, highlighting the significant role of group services in the domain of rural science and technology service. In view of resource complementarity, the group service model not only helps commissioners save on travel expenses and share knowledge and experience, but also fosters complementarity among commissioners from different professional backgrounds, further enhancing the collective synergy of collaboration. Although previous studies have focused on the impact of organizational models of agricultural extension workers on service performance, addressing factors such as influence, professional types, work stress, and competency from micro and internal perspectives [73–75], there has been a lack of discussion at the macro level about sci-tech commissioners service models. By using the Gephi social network analysis tool, our analysis not only revealed the overall service pattern of sci-tech commissioners, but also, subdivided the sci-tech commissioner groups into four distinct clusters through its built-in community detection module combined with K-means clustering analysis method. This innovative analytical approach effectively differentiates between different commissioner groups and provides more detailed evidence for a deeper understanding of the service models of sci-tech commissioners.

These four identified groups are: (1) closely-collaborating young commissioner small groups; (2) larger-scale middle-aged and young commissioner groups; (3) closely-collaborating middle-aged and senior commissioner small groups; (4) highly influential commissioners with isolated nodes.

After identifying the four groups, the research team performed an (ANOVA) to test the performance of sci-tech commissioner groups from different clusters, revealing significant differences in service performance among the clusters. There was no significant difference seen in service performance between the commissioners in Cluster 1 and those acting individually, but their performance was lower in Cluster 1 compared to that of the commissioner groups in Clusters 2, 3, and 4. The performance of the commissioner groups in Cluster 2 did not significantly differ from that of Cluster 3, but was lower than that of the commissioner groups in Cluster 4, who had the highest performance of all clusters. This finding is in alignment with existing literature which emphasizes the impact of group dynamics and organizational structure on performance outcomes [40,41]. For instance, Cluster 4's sci-tech commissioners primarily consist of "star" researchers from national-level research institutions and key universities. These organizations are missioned with delivering professional agricultural technical talents to local areas and enhancing the national agricultural technology strength overall. To this end, they undertake numerous agricultural R&D projects and have established close connections with local research institutions, government departments, businesses, and farms. The services provided by these sci-tech commissioners represent only a small part of the extensive work these "star" researchers do in agricultural technology extension, research and development, and innovation activities. Therefore, their exceptional performance could be attributed to their effective utilization of resources and close connections with key nodes within the network, both being factors considered to be crucial for the enhancement of group performance [76].

Moreover, the lack of significant difference between the performance of the commissioner groups in Cluster 1 and the individually-acting commissioners suggests that, in some cases, the average performance of group members might not exceed that of individuals

acting alone, and the average performance of the large commissioner groups may not exceed that of well-organized small commissioner groups [77,78]. Through investigation, it was found that the commissioners of Cluster 1, as well as those acting independently, are mostly staff from seed companies and county agricultural technology extension stations (typically 5–10 people in scale). Generally, no more than 5 commissioners are selected from these entities. Constrained by their entity size and geographical location, their activities are basically limited to the local county area and in point-to-point form, making it difficult to organize large-scale agricultural technology extension activities. The singleness of service mode renders less significant difference in their performances from those independently acting. Additionally, there are significant differences in the organization methods of the sci-tech commissioners between Cluster 2 and Cluster 3. Personnels in Cluster 2 are often affiliated with regional agricultural technology associations, which regularly convene members to visit key agricultural bases within the province and conduct specialized training for farmers. Therefore, the presence of some high-performance national and provincial experts in these associations significantly raises the average performance of group members. Practitioners in Cluster 3 mostly belong to national, provincial, or municipal agricultural research institutions with moderate scale (usually 20–50 people). Some institutions can select as many as 30 or more envoys, and some are even qualified to cultivate graduate students. These advantages make the envoys of these institutions very flexible in providing services, including: (1) forming small teams according to research directions (typically 2–6 people); (2) conducting large-scale trainings relying on their institutions; (3) employing various tools of agricultural technology extension, such as online teaching, video Q&A, and manual distribution. These findings prompted further exploration into the impact of different group structures and forms on the performance of technological service promotion.

To delve deeper into the core factors influencing the performance of different sci-tech commissioner groups, our research team adopted explainable ML methods which showcased the key factors affecting the classification of commissioner groups by making use of visualization techniques. Utilizing the LightGBM classification model and the SHAP value visualization, we discovered that even if two groups had similar performance levels, their group characteristics and organizational structures exhibited significant differences.

By analyzing the factors influencing the outcomes, we identified the following.

- (1) Commissioner groups in Cluster 1 are characterized by younger members with shorter work tenures and a higher number of individuals with junior titles. This is mainly caused by the low level of the dispatching departments that they belong to, the difficulty in obtaining senior professional title, the limited career development space, and the high turnover rate. These factors might lead to deficiencies in the group regarding work experience, skills in effective organization, project management, and funding support, which thereby can affect group service performance [7,8].
- (2) Commissioner groups in Cluster 2 are mainly large groups of middle-aged and young commissioners coming from a variety of different institutions, with extensive connections and complementary knowledge and skills. This is mainly because the members of the group basically belong to one or more regional agricultural associations. The diversity of group members' background means that the group can integrate the resources and advantages of all parties, form cross-field cooperation, and greatly enhance the organizing and acting capacity of the group [20]. Thanks to larger group sizes and a higher number of nodes, Cluster 2 demonstrates a higher average betweenness centrality, indicating higher efficiency as well as higher levels of cooperation among group members. In addition, as the core members of these associations are mostly composed of authoritative experts in some fields, their participation, enhances the cohesion of the organization and enhances the average service performance of the organization. A previous study has also suggested that in a network, individuals with centrality keep a group together by having a pivotal position in a network [79].
- (3) Commissioner groups in Cluster 3 consist of small groups of older, closely collaborating middle-aged and senior commissioners, with longer work tenures, and contain fewer commissioners with junior titles. Despite having fewer nodes, their tight cooperation and rich experience ensure their group performance is not inferior to that of other groups. In addition, since these institutions are usually qualified to cultivate graduate students, the mentoring mechanism of graduate education enhances the group cooperation, and on the other hand, graduate students serve as additional human resources for the promotion of agricultural technology, even though they are not counted as commissioners. Indeed, Bui et al. found that tight cooperation could impair the negative effect of group size on performance [80]. The pair-up of the apprentices and the experienced promotes the transfer of knowledge and innovation, which allows the team to provide technical services in more diverse ways [26].
- (4) Commissioner groups in Cluster 4 often undertake some major agricultural technology R&D and promotion projects, which need extensive cooperation with the government, enterprises, and farms. Therefore, in the aspect of SNA, they are characterized by nodes not belonging to any group, usually because they are part of multiple commissioner groups, having the highest eigenvector centrality values, acting as bridges and channels within the group, with connected to nodes which are also typically of higher importance. At the same time, these core nodes often enjoy high influence and trust, so that they can coordinate resources excellently and achieve technical service goals [36]. Their average performance level is the highest of all four groups, which is key to enhancing the overall group performance level. The identification of such nodes has already received attention in the academic community [81], however our current findings not only deepen our understanding of the factors affecting the performance level of science and technology commissioner groups, but also highlight the effectiveness of advanced data analysis techniques in revealing performance differences within complex networks.

5.2. Practical implications

The findings of this current study hold significant practical implications for the management of rural sci-tech commissioners. Our analysis has revealed that team-based service has become the core model of sci-tech commissioner services, noted for its advantages in

reducing service costs, promoting knowledge and experience sharing, and enhancing team collaboration and complementarity of abilities. As a result, designing an optimized team service model, establishing a group-based service mechanism, and enhancing team cohesion, connectivity, and efficiency all become critical factors which management departments must prioritize when formulating service policies for sci-tech commissioners, to ensure the team's effective and stable promotion of agricultural technology.

Optimizing the group member structure to enable more effective collaboration and complementary capabilities among sci-tech commissioners of different ages is crucial to efficiently completing agricultural technology promotion tasks. Through the adjustment of team structure, professionals can work together to better cope with challenges and improve efficiency and performance [18]. Team members with different backgrounds and different professional levels are reasonably matched to realize the exchange and sharing of knowledge and experience, making team work more innovative and diverse [19]. For example, senior commissioners should fully leverage their work experience and extensive networks by sharing their experiences through workshops or mentoring programs, thereby passing on valuable knowledge and improving the cohesion of their group. Experienced sci-tech commissioners with their deep technical knowledge, rich practical experience, and outstanding management skills, form the core strength of the team. Meanwhile, younger commissioners bring innovative vitality to the team, having proficient mastery of modern technologies and tools, as well as effective learning capabilities. Combinations such as this can effectively promote the continuous development of science and technology services and enhance the quality of services. In addition, in view of the personnel shortage in the small agricultural technology extension agencies, low professional qualifications and low institutional standards, it is suggested that local government departments adopt an innovative cooperation model, so that higher education institutions, scientific research units and grassroots extension agencies establish agricultural technology service alliances, which are responsible for the coordination of personnel activities, promoting the overall development of agricultural industry in the region. In this way, it can make full use of the scientific research advantages and human resources of universities and research institutes, while it can also improve the service ability and professional level of the commissioners of grass-roots institutions, achieving the sustainable development of regional agricultural industry.

From the perspective of information exchange and knowledge sharing, establishing a comprehensive communication and sharing mechanism to enhance mutual learning among sci-tech commissioners from different institutions is of great importance for the improvement of work capability and service efficiency. For instance, commissioners could promote the widespread dissemination of management experience and advanced technology by organizing training sessions and creating innovation platforms.

Finally, regarding the cultivation and identification of key "star" sci-tech commissioners, employing advanced technologies and methods to effectively track and support these core individuals can significantly impact and enhance connectivity and efficiency across the science and technology service network.

In summary, the findings and recommendations of this study provide both a scientific basis and practical guidance for the optimization the management of rural science and technology services. They contribute to enhancing the overall service efficacy of sci-tech commissioner teams, thereby facilitating the effective promotion of agricultural technology and technological innovation in rural development.

5.3. Limitations

This study has the following limitations, which also point to directions for future research. First, in data samples, this study is based solely on the service data of sci-tech commissioners from one province in China. Although the content and forms of SCS services across all provinces in China are broadly similar due to the guidance of centralized policies, regional differences cannot be overlooked. Therefore, future research should expand the scope of the data collection and consider the service data of sci-tech commissioners from other provinces in China to gain a more comprehensive understanding of group service models and performance levels. Second, in research methods, the analysis in this study is based on cross-sectional data drawn from Y Province in 2022, which does not capture the evolution of service modes and performance of sci-tech commissioners over time. In future studies, the research team intends to collect longitudinal data over a longer time period to better explore the developmental trends of service models and performance levels. Third, in analysis technology, this study focuses on exploring the impact of variables on service performance, paying less attention to the cumulative effect between variables as well as failing to analyze the mediating effect and moderating effect. Finally, the data-driven analysis method adopted in this study, although capable of mining information from the existing dataset, may not have fully covered some of the potential influencing factors on actual service performance, such as organizational rules and regulations, commissioner personalities, or service characteristics. Therefore, the research team's next plan is to conduct in-depth tracking studies of various sci-tech commissioner groups' service processes, and to use case analysis methods to further dissect the details and characteristics of sci-tech commissioner service models, thereby enriching and refining existing research.

6. Conclusions

Taking a social network perspective, this study explored the service performance of China's SCS by combining machine learning methods with big data mining technology. We dynamically captured group structures of science and technology services, and analyzed the factors influencing group performance. First, we categorized the commissioners into various groups and analyzed their composition and characteristics. In doing this, we found that group structure and synergy are potentially more critical to service performance than group size. Second, we examined the differences between the factors impacting the service performances of the various groups to further explain the causes behind these variations. This study provides a novel approach to assessing the service performance of the SCS in China, thereby providing support for various policymakers and departments with regards to commissioner selection processes.

Furthermore, the findings of this study can be used to improve evaluations of the service system, which can thereby help improve the overall efficacy of the SCS and other sci-tech services.

Ethics statements

Review and/or approval by an ethics committee was not needed for this study because this study did not involve any human or animal experiments.

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Data availability statement

The dataset supporting the findings of this study is available from the corresponding author upon reasonable and justified request.

CRedit authorship contribution statement

Jinghao Chen: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Wensi Li:** Data curation. **Qianxi Liu:** Writing – review & editing, Writing – original draft, Validation, Data curation. **Meihe Chen:** Data curation. **Jue Chen:** Data curation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Jinghao Chen reports financial support was provided by Chinese National Funding of Social Sciences. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix 1. Categories and Definitions of the 28 Selected Variables

Category	Variable Name	Description
Commissioner Personal Information	Commissioner Phone	Commissioner's phone number, serving as a unique identifier for the commissioner
	Gender	Commissioner's gender: male/female
	Place of Origin	Commissioner's place of origin: local/non-local
	Ethnicity	Commissioner's ethnicity: Han/minority
	Work Experience	Commissioner's years of work experience: range [1,50]
	Age	Commissioner's age: range [20,60]
	Political Affiliation	Commissioner's political affiliation: Communist Party of China/other party/non-affiliated
	Professional Title	Commissioner's professional title: junior/middle/senior/senior+/other
	Educational Background	Commissioner's educational background: technical secondary school/junior college/bachelor's degree/master's degree or above/other
	Commissioner Type	Type of commissioner: provincial/municipal/county/other
Commissioner Service Data	Service Industry	Commissioner's focus industry: planting/breeding/other
	Affiliated Institution	Name of the institution to which the commissioner belongs
	Service Days	Number of days the commissioner served, range [1275]
	Offline Training Participants	Number of participants trained by the commissioner offline, range [0,3565]
	Online Training Participants	Number of participants trained by the commissioner online: range [0,300]
	Service Briefs	Number of service briefs commissioner has submitted: range [0,180]
	Instructional Videos	Number of instructional videos commissioner has made: range [0,68]
	Promotional Materials	Number of promotional materials commissioner has distributed: range [0,33]
	Issues Resolved	Number of problems commissioner has solved for farmers: range [0,203]
	Online Replies	Number of online farmers' questions commissioner has replied to, range [0,149]
Commissioner Performance Evaluation Data	Remote Guidance	Number of remote guidance sessions commissioner has provided: range [0,165]
	Other Services	Number of other services provided by the commissioner, range [0,265]
	Satisfaction Level	Overall satisfaction level of the service recipient regarding the commissioner: satisfied/generally satisfied/dissatisfied
	Service District Scale	Size of commissioner's service district: under 100 mu/between 100 and 500 mu/over 500 mu ¹
	Service District Output Value	Output value of the commissioner's service district: within 1 million/1 to 5 million/above 5 million
	Entrepreneurial Cooperation	Whether the commissioner has started businesses, cooperatives, or other agricultural entities at the grassroots level: Yes/No

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Category	Variable Name	Description
	Media Coverage	Whether the commissioner's service has received media coverage: yes/no
	Number of Scientific Achievements	Number of scientific achievements published by the commissioner in the current year: range [0,15]

In China, the unit of land measurement is an “mu”, where 1 mu is equal to approximately 666.67 square meters or about 1/6 of an acre. For the sake of international understanding, the measurements in this paper will also be provided in hectares (ha), with 1 ha equaling 15 mu.

Appendix 2. Y Province Rural Sci-Tech Commissioners' Service Scoring Sheet

Item Type	Specific Item	Scoring Criteria	Maximum Score
Offline Services	1. Service Check-in	Days: 0.6 points per day Number of check-in, 0.2 points per occurrence	45
	2. Service District	Distance of the farthest check-in point from the same-day dispatch unit: 0.2 points for under 100 km, an additional 0.1 points for every additional 100 km Area of commissioner's district: 0 points for 100 mu or less, 1 point for between 100 and 500 mu, an additional 1 point for each additional 500 mu Estimated value of total annual output: 0 points for 1 million yuan or less, 1 point for between 100 and 500 million yuan, an additional 1 point for each additional 500 million yuan	4
	3. Technology Training	Number of check-ins with trainees: 2 points per session Number of trainee participants: 0.2 points per participant	26
Online Services	4. Live Training	Number of sessions: 2 points per online training session Online training session viewership: 0.5 point for 100 viewers or fewer, 1 point for 101 to 300 viewers, 1.5 points for 301 to 500 viewers, 2 points for over 500 viewers	5
	5. Dissemination of Information and Training Materials	Scientific achievements: 0.4 point per published item Service briefings: 0.4 point per occurrence Training videos released: 0.8 point per video released Training materials: 0.5 point per material release	5
	6. Remote Guidance	Issues resolved: 0.3 point per instance Consultations: 0.3 point per instance Remote service: 0.3 point per remote session Other remote services: 0.3 point per remote session	3
Evaluation and Assessment	7. Inspection Evaluation	Star ratings are awarded based on the service effects of the assessment district: 1 star = 5 points, 2 stars = 15 points, 3 stars = 25 points, 4 stars = 35 points, 5 stars = 45 points	45
Satisfaction	8. Service Recipient Satisfaction	Based on client ratings: Satisfied = 15 points, Basically Satisfied = 10 points, Dissatisfied = 0 points	15
Bonus Points	9. Basic Qualifications	Based on professional title: Senior = 3 points, Associate senior = 2 points, Intermediate professional = 1 point	3
	10. Number of Service Clients	0 points for 5 clients or less, 0.2 points per client from 6 onwards	2
	11. Building Entrepreneurial and Shared Interests in the Community	3 points for initiating or co-founding enterprises, cooperatives, or other business entities at the grassroots level	3
	12. Media Coverage	3 points per instance of coverage receiving by national newspapers or television	3
		2 points per instance of coverage received by provincial newspapers or television, or by national media online	2
		1 point per instance of coverage or publicity received by county-level or higher government departments official websites, city- or county-level newspapers or television, or online coverage on official dispatch unit websites at a bureau-level or higher	1
		3 points for each request to give a lecture as a representative of the technological service experience exchange	3
		Sharing key promotional materials over social media, measured by number of clicks: up to 100 people = 0.1 point, between 100 and 300 people = 0.2 point, and 300 people or more = 0.3 points	3

Appendix 3. Initial dataset for K-means clustering

Variables	Description
cluster_id	Network clique number identified by Gephi
average age	Average age of group members
group size	Number of group members
average working length	Average years of service of group members

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Variables	Description
average duration of service	Average length of service of group members
proportion of organizations with the largest number of group members	Proportion of personnel from the organization with the largest number of people in the group
total number of scientific achievements	Total number of scientific and technological achievements obtained by group members
graph density	Group graph density
average eigenvector centrality	Average eigenvector centrality of group members
average degree	Average degree of group members
average betweenness centrality	Average betweenness centrality of group members
average closeness centrality	Average closeness centrality of group members
clustering coefficient	Group clustering coefficient
number of female members	Number of female members in the group
number of male members	Number of male members in the group
number of Han members	Number of Han Chinese members in the group
number of ethnic minority members	Number of ethnic minority members in the group
number of Communist Party of China members	Number of members of the Chinese Communist Party in the group
number of other political parties' members	Number of members of other political parties in the organization
number of non-affiliated members	Number of non-affiliated members in groups
number of province-level commissioners	Number of provincial-level commissioners in groups
number of municipal-level commissioners	Number of municipal-level commissioners in the organization
number of county-level commissioners	Number of county-level commissioners in the organization
number of other types of commissioners	Number of sci-tech commissioners at other levels in the organization
number of local commissioners	Number of commissioners in the organization whose home is in the local area
number of non-local commissioners	Number of commissioners in the organization whose home is not in the local area
number of junior commissioners	Number of junior-level commissioners in the organization
number of middle commissioners	Number of commissioners with intermediate title in the organization
number of deputy senior commissioners	Number of commissioners with deputy senior title in the organization
number of senior commissioners	Number of commissioners with senior title in the organization
number of other commissioners	Number of commissioners with other titles in the organization
number of commissioners with technical secondary education	Number of commissioners with secondary school education in the organization
number of commissioners with junior college education	Number of commissioners with college degree in the group
number of commissioners with bachelor's degree	Number of commissioners with bachelor's degree in the group
number of commissioners with master's degrees	Number of commissioners with master's degree in the group
number of commissioners with other educational backgrounds	Number of commissioners with other educational backgrounds in the group
number of commissioners serving plating industries	Number of commissioners working in the planting industry in the group
number of commissioners serving breeding industries	Number of commissioners working in the breeding industry in the group
number of commissioners serving other industries	Number of commissioners in other industries in the group

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